Classical Field Theory — Exercise Sheet 17

Forced and accidental massless directions

Hints and answers

Use the real scalar kinetic coefficient $1/2$ and the orbit–Hessian lemma of Lecture 17. Counts of massless modes below refer to the linearised field. A vacuum background need not belong to L^2; perturbation data may be Schwartz or in the earlier energy class.

Exercise 17.1 (The full orthogonal orbit). For $N \geq 3$, $\lambda, v > 0$, consider

$$V(u) = \frac{\lambda}{8}(|u|^2 - v^2)^2, \qquad u \in \mathbb{R}^N, \qquad u_0 = vn, \quad |n| = 1,$$

with the usual $O(N)$ action and canonical kinetic term. Find the vacuum set, the stabiliser at u_0, its dimension, and the orbit tangent. Derive the Hessian and diagonalise the linearised field using the radial and tangential projectors, without first rotating n to a coordinate axis. Identify which masses are symmetry-forced. Finally discuss $N = 1$ separately: does its broken discrete symmetry force a massless mode?

Hint. An orthogonal transformation fixing n restricts to an arbitrary orthogonal transformation on $n^\perp$. Compute the gradient before substituting the vacuum.

Answer. The stabiliser is $O(N-1)$, the orbit is S^{N-1}_v, and $D^2V(u_0) = \lambda v^2 nn^{\mathsf{T}}$. There is one mass $\sqrt{\lambda}v$ and $N-1$ forced massless components. At $N = 1$ there are no tangent modes and the mass remains $\sqrt{\lambda}v$.

Exercise 17.2 (A flat singlet still has a nonlinear restoring force). Add a real scalar z with its own canonical kinetic term and potential

$$\widetilde{V}(u, z) = \frac{\lambda}{8}(|u|^2 - v^2)^2 + z^4.$$

Let $O(N)$ act on u and trivially on z. Find the vacuum set, the Hessian kernel at $(vn, 0)$, and the subspace forced by the symmetry. Can the extra singlet direction be tangent to a curve of vacua? On the spatial torus, set $u = vn$ and take a homogeneous singlet $z(0) = A > 0$, $z_t(0) = 0$. Derive its nonlinear period as a definite integral and its dependence on A. Explain how a zero linearised mass is compatible with this nonlinear oscillation. These homogeneous data are not claimed to be L^2 data on $\mathbb{R}^d$.

Hint. The nonnegative summands must both vanish at a minimum. Use the conserved mechanical energy $z_t^2/2 + z^4$ and integrate one quarter of the periodic orbit between zero and a turning point.

Answer. The vacuum set is $S^{N-1}_v \times \{0\}$; the Hessian kernel has dimension N, but only $N-1$ directions are symmetry-forced. The singlet period is

$$T_A = \frac{2\sqrt{2}}{A} \int_0^1 \frac{\mathrm{d}s}{\sqrt{1 - s^4}}.$$

Exercise 17.3 (An ineffective group factor belongs to the stabiliser). Take $G = SO(3)\times U(1)$ acting on a real triplet by $(R, e^{i\theta})u = Ru$. Use the radial potential of Problem 1 with $N = 3$ and vacuum $u_0 = v(2,-1,2)/3$. Determine the full stabiliser and the differential of the orbit map $(c,\theta) \mapsto c \times u_0$ on $\mathbb{R}^3 \oplus \mathbb{R}$. Find a basis of its kernel, the number of broken directions and the Hessian nullity. Why would counting $\dim G$ or only counting rotations that fix u_0 give a wrong answer?

Hint. The entire $U(1)$ factor acts trivially. A rotation generator fixes a nonzero vector exactly when its axial vector is parallel to that vector.

Answer. $H \cong SO(2) \times U(1)$ has dimension two, while $\dim G = 4$. The kernel is spanned by $((2,-1,2),0)$ and $(0,1)$, where the latter has zero rotation part. There are two broken and two massless directions.

Exercise 17.4 (Kinetic normalisation before diagonalising masses). For a real three-component perturbation h, consider the quadratic density

$$L = \frac{1}{2}\partial_\mu h^{\mathsf{T}} K \partial^\mu h - \frac{1}{2} h^{\mathsf{T}} M h, \qquad K = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad M = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 9 & 9 \\ 0 & 9 & 9 \end{pmatrix}.$$

The real phase data may be taken Schwartz. Verify the factorisation $K = S^{\mathsf{T}} S$ for

$$S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Derive the canonical momentum, and find the corresponding change of coordinates and momenta that preserves $\int \delta h_a \wedge \delta p_a$. Compute the physical squared masses and compare them with the raw eigenvalues of M. State the general eigenproblem and explain what happens to the kernel dimension under this kinetic normalisation.

Hint. Use $Q = Sh$ and solve for the new momentum from $p^{\mathsf{T}}\delta h$. Transform the potential by congruence, not by similarity of M alone.

Answer.

$$p = K h_t, \qquad Q = Sh, \qquad P = S^{-\mathsf{T}} p, \qquad S^{-\mathsf{T}} M S^{-1} = \operatorname{diag}(0,4,9).$$

The squared masses are $0, 4, 9$; the raw Hessian eigenvalues are $0, 4, 18$.